Notes: Coppola (RFS, 2025)

In Safe Hands: The Financial and Real Impact of Investor Composition over the Credit Cycle

Contents

1 Big picture

- **Question.** Does the composition of bondholders matter for bond price dynamics during crises, and do those pricing effects spill over to firms' financing and investment?
- **Setting.** The baseline laboratory is the U.S. investment-grade corporate bond market from 2005 to 2020, combining NAIC insurer holdings, Morningstar mutual fund and ETF holdings, and transaction-level prices from TRACE. The analysis then extends to the EMU, the United Kingdom, and Canada.
- **Core empirical object.** The key state variable is the insurer share in a bond, compared with ownership by open-end funds. The main outcome is the bond's crisis *drawdown*, defined from excess returns relative to duration-matched Treasuries.
- **Main pricing result.** Holding issuer and bond characteristics fixed, increasing insurer ownership by 50 percentage points causes crisis drawdowns to be about 20% shallower. In other words, safe-handed ownership materially reduces fire-sale losses.
- **Main real-side result.** Firms with more insurer-held debt before the Great Recession raise more new bond financing during the crisis, pay lower offering yields, and invest more. Moving from the 10th to the 90th percentile of pre-crisis insurer share raises capital expenditures by about 1.5 percentage points of assets.
- **Mechanism.** Mutual funds have redeemable, open-end liabilities and face procyclical investor withdrawals, so they sell into bad times. Insurers have stickier liabilities and surrender restrictions, so they are much less exposed to forced selling. Ownership composition therefore maps into state-contingent price pressure.
- **Why identification is plausible.** The paper compares near-identical bonds issued by the same firm and also exploits quasi-random primary-market allocations to large insurers via a shift-share instrument. The core idea is that insurers buy mostly at issuance, hold bonds for a long time, and each large insurer is only allocated a subset of new issues.
- **Why the paper matters.** The paper links portfolio-balance ideas to fire-sale macro-finance. It shows that ownership structure is not just a pricing detail: it affects firms' cost of capital and thus capital allocation during downturns.

2 Data and measurement (key objects)

2.1 Institutional holdings data

- **Insurers.** U.S. insurer positions come from quarterly Schedule D filings to the NAIC. These data cover the full security holdings of U.S. insurers.
- **Funds.** Mutual fund and ETF holdings come from Morningstar and cover both U.S.-domiciled and foreign-domiciled funds.
- **Scale of the data.** By 2020, the combined security-level holdings data cover nearly $60 trillion in global wealth.
- **Concentration.** In 2015, the insurer sample contains 1,225 firms with mean invested assets of $3.51 billion, and the top 50 insurers hold 71% of sector assets. The U.S. fixed-income specialist

fund sample contains 1,957 funds with mean assets of $1.67 billion, and the top 50 hold 44% of sector assets.

- **Why concentration matters.** The insurer size distribution is fat-tailed, so idiosyncratic purchase decisions by a few large insurers survive aggregation and become usable identifying variation.

2.2 Bond-level pricing and sample construction

- **U.S. prices.** The baseline uses transacted corporate bond prices from TRACE, after standard data-cleaning procedures.
- **Non-U.S. prices.** Outside the United States, the paper uses S&P Global prices, traded when available and estimated otherwise.
- **Sample.** The main U.S. sample is actively traded investment-grade corporate bonds. The number of outstanding bonds rises from 7,951 in 2007 to 11,293 in 2020; average issue size rises from $407 million to $698 million.
- **Why only investment-grade bonds?** Insurers hold very little high-yield debt because risk-based capital charges make those bonds unattractive. In 2015, only 5% of insurer corporate-bond holdings are high yield versus 37% for U.S. fixed-income funds.

2.3 Key ownership and bond-characteristic facts

- **Coverage of the U.S. market.** By 2020, insurers hold 35% of the U.S. investment-grade market in the microdata, U.S. funds hold another 22%, and foreign funds hold 9%. Altogether, the observable holdings cover about two-thirds of the whole market.
- **Secular shift.** The insurer share falls from 52% in 2007 to 35% by 2020, while the role of funds rises steadily. This makes the paper's fragility mechanism quantitatively important.
- **Portfolio characteristics.** Insurers hold longer-duration bonds than U.S. funds: in 2015, average duration is 7.0 years for insurers versus 5.5 for U.S. specialist funds.
- **Similarities and differences.** Issue size, seniority, and callability look broadly comparable across sectors, but duration and credit-risk composition differ enough that very rich fixed effects are needed.

2.4 Main variables

- **Excess return.** Weekly bond returns are measured relative to a duration-matched basket of Treasuries, so the paper isolates credit-spread movements rather than risk-free rate movements.
- **Drawdown.** For each crisis event, the paper computes the maximum drop in the cumulative excess-return index relative to the event start.
- **Insurer share.** Bond-level insurer ownership is measured as the market value of insurer holdings divided by the bond's market value outstanding.
- **Proxy safe-hand share abroad.** In non-U.S. markets, the safe-hand share is proxied by one minus the fund share, because holdings of insurers and pension funds are not directly observed.

- **Firm-level treatment.** For the real-outcomes analysis, the paper aggregates insurer ownership across a firm's outstanding bonds as of 2007Q1.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Excess returns, cumulative returns, and drawdowns

The weekly excess return on bond i is

$$R^e_{i,t} = R_{i,t} - R^f_t(D_{i,t}), \tag{1}$$

where $R_{i,t}$ is the raw weekly net return on the corporate bond and $R^f_t(D_{i,t})$ is the return on a basket of Treasury bonds with matched duration $D_{i,t}$.

The cumulative excess-return index since the bond's first trading date t^0_i is

$$CR_{i,t} = \prod_{\tau=t^0_i}^{t} \left(1 + R^e_{i,\tau}\right). \tag{2}$$

The insurer share in bond i at time t is

$$\phi_{i,t} = \frac{X^I_{i,t}}{V_{i,t}}, \tag{3}$$

where $X^I_{i,t}$ is the market value of insurer holdings and $V_{i,t}$ is the bond's market value outstanding.

For event window $T_e = (t^S_e, \dots, t^E_e)$, the crisis drawdown is

$$\zeta_{i,e} = \max_{\tau \in T_e} \left(1 - \frac{CR_{i,\tau}}{CR_{i,t^S_e}}\right). \tag{4}$$

- **Interpretation.** These equations convert raw prices into a clean crisis-loss object. The left-hand side of the main regression is not a level or spread but the worst cumulative credit loss within the event window.
- **Why this construction matters.** By using excess returns and duration-matched Treasuries, the paper strips out pure interest-rate movements and isolates credit-market stress.
- **Magnitude.** A drawdown of 0.40 means that the cumulative excess-return index falls to 60% of its event-start value at the trough.

3.2 Baseline within-firm crisis-pricing regression

The core estimating equation is

$$\log \zeta_{i,e} = \alpha + \beta\, \phi_{i,t^S_e - 12} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}. \tag{5}$$

- **Dependent variable.** The log drawdown of bond i in crisis episode e.

- **Regressor of interest.** The insurer share measured in the quarter just before the event begins.
- **Fixed effects.** The saturated specification interacts event dummies with issuer dummies and bond characteristics: rating, duration bucket, issue-size decile, callability, seniority, and coupon type.
- **Interpretation of β.** This is the semielasticity of crisis losses with respect to insurer ownership. The most saturated estimate is $\hat{\beta} = -0.41$.
- **Economic magnitude.** Raising insurer ownership by 50 percentage points makes drawdowns about 20% shallower.

3.3 Residualized variation inside the fixed effects

To show that the identification is not coming from coarse cross-sectional differences, the paper residualizes both drawdowns and ownership against the saturated fixed effects:

$$\zeta_{i,e} = \alpha + \text{Interacted Fixed Effects} + \nu^{\zeta}_{i,e}, \tag{6}$$

$$\phi_{i,t^S_e-12} = \alpha + \text{Interacted Fixed Effects} + \nu^{\phi}_{i,e}. \tag{7}$$

- **Interpretation.** The residual histograms in Figure 6 show that there is still substantial within-issuer, within-characteristics dispersion in both ownership and outcomes.
- **Why it matters.** The main result is not driven by insurers simply holding safer firms or bonds with obviously different observables.

3.4 Persistence of primary-market ownership and post-issuance liquidity

The persistence of ownership decisions made at issuance is estimated from

$$\phi_{i,\tau(i)+12s} = \alpha_s + \lambda_s\, \phi_{i,\tau(i)} + \varepsilon_{i,s}, \tag{8}$$

where $\tau(i)$ is the issuance date and one unit of s corresponds to one quarter.
Post-issuance liquidity decay is measured using

$$T_{i,\tau(i)+12s} = \alpha_s + \lambda_s\, T_{i,\tau(i)} + \varepsilon_{i,s}, \tag{9}$$

where $T_{i,t}$ is either quarterly trading volume or the number of trades.

- **Interpretation.** These regressions show why initial allocations matter. Insurer ownership chosen at issuance remains highly persistent for years, while bond-market liquidity collapses sharply after the issue quarter.
- **Key findings.** Ten years after issuance, initial insurer ownership still explains about 40% of contemporaneous ownership. One quarter after issuance, trading volume is only about 10% of issuance-quarter volume.
- **Economic idea.** Insurers acquire large positions when the whole issue is available in the primary market and then rarely rebalance afterward.

3.5 Decomposition of ownership and the shift-share instrument

The paper decomposes insurer ownership as

$$\phi_{i,t} = \sum_{j \in \mathcal{J}} \xi_{i,j}\, \tilde{\phi}_{i,j,\tau(i)} + \hat{\phi}_{i,t}, \tag{10}$$

where $\xi_{i,j}$ indicates whether insurer j bought bond i at issuance, $\tilde{\phi}_{i,j,\tau(i)}$ is the position size conditional on purchase, and $\hat{\phi}_{i,t}$ captures later secondary-market trading.

The security-level shift-share instrument is

$$Z_i = \sum_{j \in \mathcal{J}} \xi_{i,j}\, \bar{\phi}_j, \tag{11}$$

where $\bar{\phi}_j$ is insurer j's average share of aggregate insurer primary-market purchases.

A sufficient exclusion condition is

$$\mathbb{E}[\xi_{i,j}\, \varepsilon_{i,e} \mid \mathcal{F}_{i,e}] = 0 \quad \text{for all } i, j. \tag{12}$$

- **Interpretation.** The instrument keeps only the extensive-margin part of primary-market allocation: which insurers happened to buy the bond at issuance.
- **Relevance.** The first stage is strong because large insurers buy mostly at issuance and then keep those bonds. In Table 4, the first-stage F-statistic is about 72 to 73.
- **Exclusion restriction.** Conditional on issuer and bond observables, the particular matching of insurers to nearly identical bonds in the primary market must be unrelated to future crisis performance.
- **Key result.** The 2SLS estimate in the saturated specification is $\hat{\beta} = -0.45$, close to the OLS estimate of -0.41.

3.6 Proxy specification for non-U.S. markets

For non-U.S. bonds, the paper defines the proxy safe-hand share as

$$\tilde{\phi}_{i,t} = 1 - \frac{X^F_{i,t}}{V_{i,t}}, \tag{13}$$

and estimates

$$\log \zeta_{i,e} = \alpha + \beta\, \tilde{\phi}_{i,t^S_e - 12} + \rho_{c(i)} + \text{Interacted Fixed Effects} + \varepsilon_{i,e}. \tag{14}$$

- **Interpretation.** Since foreign insurer and pension holdings are not directly observed, the complement of fund holdings proxies for stable owners.
- **Why the currency fixed effects?** Outside the U.S., bonds are denominated in multiple currencies, so the regression controls for denomination-specific spread behavior.
- **Result.** The pooled non-U.S. estimate is -0.51, with similar negative estimates in the EMU, U.K., and Canadian samples.

3.7 Firm-level aggregation and real outcomes

The firm-level insurer share before the crisis is

$$\Phi_k = \frac{\sum_{i \in \mathcal{I}_k} X^I_{i,2007Q1}}{\sum_{i \in \mathcal{I}_k} V_{i,2007Q1}}. \tag{15}$$

The firm-level instrument aggregates the security-level instrument:

$$\mathcal{Z}_k = \sum_{i \in \mathcal{I}_k} \rho_{i,k} Z_i, \tag{16}$$

with bond-size weights $\rho_{i,k}$.

The dynamic difference-in-differences specification for investment is

$$\text{Investment}_{k,t} = \alpha_k + \gamma_t + \sum_{\tau} \beta_\tau \, \Phi_k \, \mathbf{1}\{t = \tau\} + \text{Controls} + \varepsilon_{k,t}. \tag{17}$$

A static version that emphasizes rollover risk is

$$\text{Investment}_{k,t} = \alpha_k + \gamma_t + \beta \, \Phi_k \cdot \text{Post}_t + \eta \, \Phi_k \cdot \text{Post}_t \cdot M_k + \text{Controls} + \varepsilon_{k,t}, \tag{18}$$

where M_k is the share of outstanding debt due in 2008 or 2009.

For credit outcomes, the paper estimates

$$Y_{k,t} = \alpha_k + \gamma_t + \sum_{\tau} \beta_\tau \, \Phi_k \, \mathbf{1}\{t = \tau\} + \text{Controls} + \varepsilon_{k,t}, \tag{19}$$

where $Y_{k,t}$ is either a new-issuance indicator or the offering yield on newly issued bonds.

- **Interpretation.** These specifications ask whether firms financed by safe-handed bondholders are better able to preserve borrowing and investment during the Great Recession.
- **Identification.** The paper instruments the interaction terms using the firm-level shift-share instrument interacted with year dummies.
- **Key results.** A 50-point increase in Φ_k raises CAPX by about 1.5 percentage points of assets, raises CAPX plus acquisitions by about 2 to 3 points, increases the probability of issuing new bonds by 25 to 30 percentage points in 2008–2009, and lowers offering yields by roughly 120 basis points.
- **Rollover channel.** The effect is stronger for firms with more debt maturing during the crisis.

3.8 Relationship channel in the primary market

For new bond issue i by firm k at time t, the paper estimates

$$\xi_{i,j,k,t} = \eta_t + \alpha_k + \gamma \, \mathcal{R}_{i,j,k,t} + \sum_{s} \beta_s \, \mathcal{R}_{i,j,k,t} D_{s,k,t} + \sum_{s} \rho_s D_{s,k,t} + \varepsilon_{i,j,k,t}, \tag{20}$$

where $\xi_{i,j,k,t}$ indicates whether insurer j buys the new issue, $\mathcal{R}_{i,j,k,t}$ indicates whether that insurer already holds another bond by firm k, and $D_{s,k,t}$ are secondary-market spread deciles for the issuer.

- **Interpretation.** The coefficient γ captures the baseline relationship effect, while the sequence β_s shows how that effect changes when a firm's existing bonds are already trading at wider spreads.
- **Result.** Existing insurer relationships raise purchase probability by 5.5 percentage points even for low-spread firms, and the effect rises by nearly another 8 percentage points from the bottom to the top spread decile.
- **Economic meaning.** Safe-handed owners stabilize not just secondary-market prices but also access to new credit.

3.9 Pricing outside crises

The normal-times pricing regression is

$$s_{i,t} = \alpha + \bar{\Lambda}\,\phi_{i,t-12} + \tilde{\Lambda}\,\phi_{i,t-12}\mathrm{GZ}_t + \text{Interacted Fixed Effects} + \varepsilon_{i,t}, \tag{21}$$

where $s_{i,t}$ is the bond's yield spread to Treasury in basis points.

- **Interpretation.** The total pricing effect is $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda}\mathrm{GZ}_t$, so the paper allows ownership effects to vary smoothly with aggregate credit stress.
- **Key fact.** Outside crises, bonds with higher insurer shares have slightly *higher* yield spreads—about 20 basis points when ownership moves from 0 to 1. During crises the sign flips because those bonds experience smaller fire-sale losses.
- **Economic idea.** Holding bonds with insurers looks like buying insurance: firms pay a small liquidity premium in good times but receive more stable financing in bad times.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline within-firm identification

- **Comparison.** The baseline design only compares bonds issued by the same legal entity and with nearly identical observable characteristics.
- **What gets absorbed.** Issuer fixed effects remove firm-level differences in resilience, while interacted bond-characteristic fixed effects remove systematic differences by rating, maturity, size, seniority, coupon type, and callability.
- **Intuition.** The design compares one bond held more by insurers and another held more by funds, even though both are essentially the same claim on the same borrower.
- **Concrete example.** The paper discusses two very similar American Express Credit Corporation bonds: pre-crisis insurer shares are 22% versus 4%, and the Great-Recession drawdowns are 32% versus 51%.

4.2 Why insurer ownership varies within issuer

- **Primary-market allocations.** Large insurers typically enter at issuance and buy large blocks when underwriters allocate the deal.
- **Idiosyncrasy.** Even very large insurers only buy a subset of available bonds. The paper shows that none of the 300 largest insurers holds more than 61% of the top 500 investment-grade bonds, whereas many insurers effectively hold the full S&P 500 in equities.
- **Persistence.** Those issuance decisions remain in the data for years because insurers do not trade much afterward.
- **Aggregation.** Because the insurer size distribution is very concentrated, those idiosyncratic decisions do not wash out in aggregate.

4.3 Shift-share IV logic

- **First-stage source of variation.** Whether a bond is allocated to large or small insurers at issuance shifts the aggregate insurer share of that bond.
- **What the instrument strips out.** It removes potentially endogenous secondary-market trading and also strips out intensive-margin changes coming from aggregate insurer balance-sheet variation.
- **Exclusion concern.** The main threat is that larger insurers systematically obtain the bonds that would have drawn down less anyway for unobserved reasons.
- **Why the author argues against that concern.** Insurers are benchmark-style investors, not tactical within-firm security selectors, and a bootstrap test in the appendix rejects residual alpha in the cross-section of insurers.

4.4 Partial-identification argument

- **Oster-style robustness.** The paper computes how strong selection on unobservables would have to be, relative to selection on observables, to wipe out the baseline result.
- **Threshold.** The required ratio is $\bar{\delta} = 3.2$.
- **Interpretation.** Unobservables would need to be more than three times as important as the rich included observables to make the treatment effect zero, which is strong evidence against omitted-variable stories.

4.5 Firm-level caveat

- **Security versus firm level.** The real-outcomes regressions aggregate the instrument to the firm level and therefore require stronger assumptions than the security-level pricing tests.
- **Why still plausible.** The cross-section of bonds outstanding per firm is fairly small, so a lot of the security-level variation survives the aggregation. The standard deviation of the firm-level instrument is still 76% of the security-level one.
- **Inference note.** Security-level regressions cluster at the issuer-by-event level; firm-level regressions cluster at the firm level; the relationship regression clusters at the issuer level.

5 Tables and figures

5.1 Table 1: Holdings data summary

Read:

- The insurer sector is much more concentrated than the fund sector. In 2015, the top 50 insurers hold 71% of insurer assets, while the top 50 U.S. fixed-income specialist funds hold 44%.
- That concentration is central for the shift-share design: allocations to a few large insurers move aggregate insurer ownership a lot.
- The table also shows the immense scale of the holdings data and justifies treating insurers and funds as the dominant investor sectors in the market.

Table 1
Holdings data: Sample summary statistics

		Total invested assets ($M)				Concentration ratios		
	N	Mean	Median	95th percentile	SD	CR_{10}	CR_{50}	CR_{100}
***A.* U.S. insurers**								
2005	1,208	2,804	110	10,914	12,616	0.34	0.73	0.85
2010	1,242	3,048	118	12,397	13,994	0.34	0.72	0.86
2015	1,225	3,510	138	14,770	15,399	0.34	0.71	0.85
2020	1,216	4,396	163	19,813	19,821	0.35	0.70	0.85
***B.* U.S. funds (all)**								
2005	5,597	1,080	219	3,816	4,948	0.14	0.32	0.42
2010	6,675	1,570	267	6,000	6,587	0.11	0.29	0.39
2015	8,193	1,948	291	7,167	8,546	0.11	0.27	0.37
2020	8,418	3,715	404	12,768	23,616	0.16	0.34	0.44
***C.* Non-U.S. funds (all)**								
2005	11,135	681	97	1,989	5,303	0.18	0.38	0.46
2010	20,135	580	81	1,676	9,360	0.22	0.33	0.39
2015	22,069	878	118	2,449	8,563	0.18	0.33	0.40
2020	25,152	1,136	151	3,441	10,209	0.13	0.25	0.33
***D.* U.S. funds (specialists)**								
2005	1,565	794	196	2,839	4,968	0.28	0.46	0.58
2010	1,529	1,503	307	6,160	7,243	0.26	0.48	0.61
2015	1,957	1,667	324	6,236	6,206	0.21	0.44	0.56
2020	2,097	2,881	464	11,197	11,666	0.21	0.45	0.58
***E.* Non-U.S. funds (specialists)**								
2005	2,506	574	150	2,454	1,527	0.12	0.29	0.42
2010	4,150	522	119	2,135	1,624	0.11	0.27	0.38
2015	5,090	687	149	2,744	2,481	0.12	0.26	0.36
2020	6,087	980	208	3,716	3,884	0.12	0.27	0.36

This table shows the total number of insurers as well as mutual funds and exchange traded funds (ETFs) in the sample. I report statistics separately for all funds and for fixed-income specialist funds (as determined via Morningstar's categorization). I also show statistics on total invested assets (equities and bonds), as well as concentration ratios detailing the share of total invested assets held by largest 10, 50, and 100 institutions in each group. Insurers and funds with less than $25 million in total invested assets are excluded. Data shown for the years 2005, 2010, 2015, and 2020.

Table 2
Overview of corporate bond holdings of insurers and fixed-income specialist funds

	Average duration (years)	Average issue size ($M)	Share of holdings in category (%)			
			High yield	Junior	Floating	Callable
***A.* U.S. insurers**						
2005	6.2	563	8	8	3	57
2010	6.4	678	6	5	2	70
2015	7.0	726	5	4	1	78
2020	8.1	785	4	4	4	77
***B.* U.S. funds[a]**						
2005	4.9	556	55	4	9	66
2010	5.1	733	40	5	7	61
2015	5.5	761	37	6	7	71
2020	5.9	814	27	6	13	77
***C.* Non-U.S. funds[a]**						
2005	5.7	615	18	9	19	24
2010	4.7	688	28	7	17	28
2015	5.3	772	36	10	15	32
2020	5.4	808	30	11	21	45

a*Fixed-income specialists only*
This table reports summary statistics on the characteristics of the corporate bond holdings of insurance companies and fixed-income specialist funds. These include average bond duration, the share of holdings that is in high-yield bonds, the share of holdings that is in junior bonds, the share of holdings that is in floating-rate bonds, the share of holdings that is in bonds with callable features, and the average issue size for bonds held.

5.2 Table 2: Characteristics of insurer and fund bond portfolios

Read:

- Insurers hold somewhat longer-duration bonds than U.S. funds and dramatically fewer high-yield bonds.
- These differences show why naive cross-sectional comparisons would be biased.
- The table motivates the saturated fixed-effect structure in the baseline regressions.

5.3 Figure 1: Market coverage and changing investor shares

Read:

- The figure shows that the microdata capture roughly two-thirds of the U.S. investment-grade market by 2020.
- It also shows a secular shift from insurers toward funds. This makes the ownership-composition channel increasingly important over time.

- The empirical setting is therefore both high-coverage and macro-relevant.

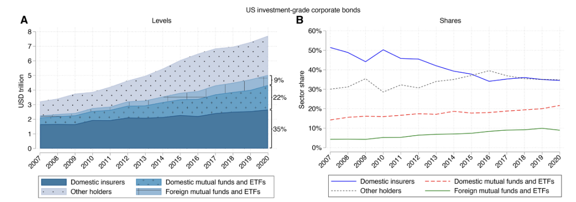

Figure 1
Aggregate value of U.S. investment-grade corporate bonds and market coverage via security-level holdings micro-data
The total shaded region in panel A represents the aggregate outstanding value of U.S. investment-grade corporate bonds in the sample, defined as bonds that are reportable in TRACE. The *dark blue area* represents the aggregate value of the positions in these bonds held by U.S. insurers, as reported in the NAIC data. The remaining shaded areas of panel A represent the value of the positions in these bonds held by investment funds (mutual funds and ETFs) in the Morningstar data, split according to whether or not the funds are U.S. domiciled. Panel B shows the corresponding shares of the aggregate value that each sector accounts for, providing a counterpart to the picture in levels of panel A.

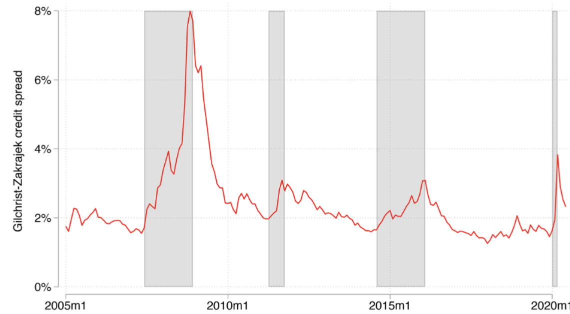

Figure 2
Events of focus
The *solid red line* plots the Gilchrist and Zakrajšek (2012) index, which provides an aggregate measure of credit spreads on U.S. corporate bonds. The *shaded gray areas* show the event windows used to estimate bond drawdowns in my analysis. The start and end dates for each event are defined mechanically so that they correspond to the prior trough and subsequent peak values in the GZ index. Section 2.2 discusses the drivers of aggregate credit spread variation in these episodes.

5.4 Figure 2: Crisis windows

Read:

- The paper studies four major stress episodes: the Great Recession, the 2011 sovereign-downgrade episode, the 2014–2016 commodity slump, and the March 2020 COVID crash.
- The windows are defined mechanically from trough-to-peak movements in the GZ credit-spread index.
- This makes the drawdown analysis systematic rather than ad hoc.

5.5 Table 3: Baseline OLS estimates

Read:

- The coefficient on insurer share remains stable as the fixed effects get more saturated, ending at -0.41 in the fully saturated specification.
- That stability is important: the result does not disappear once issuer and bond observables are fully controlled for.
- The economic interpretation is large—a 50-point increase in insurer share implies roughly 20% shallower drawdowns.

5.6 Figure 3: Robustness

Read:

- The baseline estimate survives stricter duration controls, bond-age controls, Rule 144A controls, underwriter controls, covenant controls, and multiple sample restrictions.
- The estimate is also robust to excluding AIG, excluding callable bonds, excluding financials and utilities, and controlling for noncrisis betas.
- The broad message is that the coefficient is not being driven by one narrow sample choice.

Table 3
Investor base effects for U.S. corporate bonds: Baseline OLS estimates

	Logarithmic bond drawdown				
	(1)	(2)	(3)	(4)	(5)
Insurer share	−.43	−.60	−.46	−.40	−.41
	(.06)	(.05)	(.05)	(.10)	(.10)
Fixed effects interaction:					
Event dummy	Yes	Yes	Yes	Yes	Yes
Bond size	—	Yes	Yes	Yes	Yes
Bond duration	—	Yes	Yes	Yes	Yes
Bond rating	—	—	Yes	Yes	Yes
Issuer dummy	—	—	—	Yes	Yes
Callable dummy	—	—	—	—	Yes
Senior dummy	—	—	—	—	Yes
Coupon type	—	—	—	—	Yes
Identifying observations	15,012	14,996	14,649	5,328	4,945
Identifying issuers	1,430	1,430	1,412	452	437
R^2	.29	.60	.68	.84	.85

This table reports the estimates of the regression specification in Equation (5) in the sample of actively traded investment-grade U.S. corporate bonds. The specification in column 1 includes event fixed effects. Column 2 includes interacted event by bond size decile by duration category fixed effects. Columns 3 and 4 add additional interactions with rating and issuer dummies. Column 5 includes the fully-saturated set of interacted fixed effects, which include all of the above variables plus dummies for bond seniority, bond callability, and coupon type. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

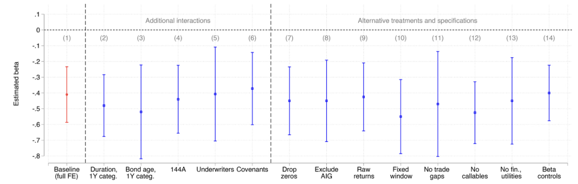

Figure 3
Investor base effects for U.S. corporate bonds: Robustness
This figure demonstrates the robustness of the baseline estimate (*red bar*) from column 5 of Table 3 to further variations in the empirical specification. Bars 2 through 4 introduce further interactions to the fully-saturated set of fixed effects. Bar 2 introduces an additional interaction with stricter duration categories, with duration rounded to the nearest year. Bar 3 adds an interaction with bond age, rounded to the nearest year. Bar 4 includes an additional interaction with a dummy indicating whether a bond is issued under SEC rule 144A. Bar 5 adds an interaction with dummies for the set of a bond's lead underwriters. Bar 6 adds an interaction with dummies for the set of a bond's covenants, hence removing bonds with nonmatching covenants from the estimation sample. Additionally, bars 7 through 14 consider alternative sample treatments and specifications. Bar 7 excludes bonds with a zero insurer share from the sample. Bar 8 excludes the positions of AIG from the computation of the insurer positions $X^I_{i,t}$. Bar 9 uses raw returns $R_{i,t}$ in place of the excess returns $R^e_{i,t}$. Bar 10 uses an alternative measure of drawdowns which applies the same window to all bonds for a given event. Bar 11 drops any bonds that trade less frequently than once a week in a given event period. Bar 12 excludes callable bonds. Bar 13 excludes financials and utilities. Bar 14 adds noncrisis betas as an additional control. The displayed confidence intervals are at the 95% confidence level and are clustered at the issuer by event level.

5.7 Figures 4 and 5: Mechanism through liability structure

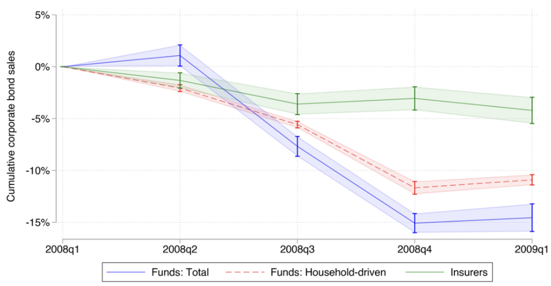

Figure 4
Corporate bond sales dynamics in the Great Recession
The *solid green line* plots the cumulative change in corporate bond holdings for the asset-weighted average insurer in the sample over the course of the Great Recession, relative to their level in 2008Q1. The changes in corporate bond holdings are measured using par values, so that they reflect pure quantity changes rather than any valuation effects owing to variation in market prices. The *solid blue line* shows the equivalent portfolio dynamics for the asset-weighted average mutual fund or ETF in the sample. The *dashed red line* plots the component of the cumulative net corporate bond sales by funds that is household-driven, as defined in Section 2.4. The vertical bars show 95% confidence intervals around these estimates.

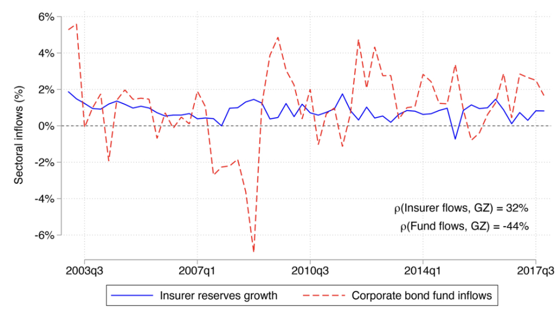

Figure 5
Flows in and out of insurers and fixed-income funds
The *solid blue line* represents the quarterly net change in policy reserves for the aggregate insurance sector, scaled by lagged total policy reserves. The *dashed red line* represents the quarterly value of aggregate net client inflows and outflows for fixed-income mutual funds and ETFs in the sample, scaled by lagged total assets under management. The *black annotations* report the correlation of these series with the quarterly GZ spread index.

Read:

- Figure 4 shows the quantity mechanism: by 2009Q1 insurers reduce corporate-bond holdings by less than 5%, while funds reduce them by 15%.
- More than two-thirds of fund net sales are household-driven rather than discretionary portfolio management.
- Figure 5 shows why: fund inflows are strongly procyclical (correlation with the GZ index $= -44\%$), while insurer reserve growth is countercyclical (correlation $= +32\%$).

5.8 Figures 6 through 9: Why the instrument works

Read:

- Figure 6 shows that after residualizing by the saturated fixed effects there is still substantial within-cell dispersion in ownership and drawdowns.
- Figure 7 shows idiosyncratic bond holdings: insurers do not come close to holding the full bond market portfolio, unlike in equities.

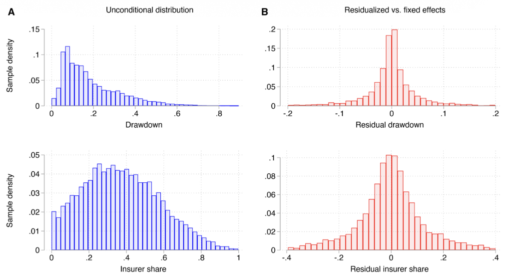

Figure 6
Residual variation within the fixed effects
The *blue histograms* in panel A plot the sample distribution of drawdowns and insurer shares in the sample of actively traded U.S. investment-grade corporate bonds that underlies the regression estimates in Table 3. The *red histograms* in panel B plot the sample distribution of the same quantities after residualizing them against the fully-saturated set of interacted fixed effects.

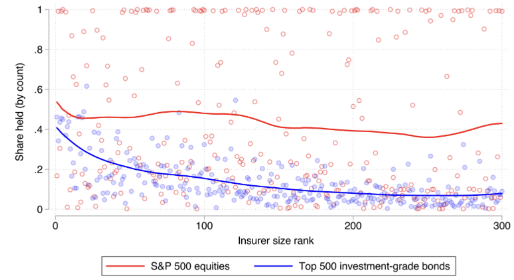

Figure 7
Idiosyncrasy in insurers' corporate bond holdings at the security level
This figure ranks insurers by the overall size of their portfolio investment, and shows the share of the 500 largest actively traded investment-grade U.S. corporate bonds that they hold an open position in (*blue dots*) as well as the share of S&P 500 equities that they hold an open position in (*red dots*), all measured cross-sectionally as of 2017Q4. Indirect equity positions held through mutual funds and ETFs are included. The thick lines display the fit from LOWESS regressions of the investment shares (shown by the dots) on the insurer rank.

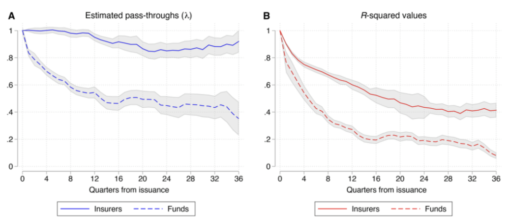

Figure 8
Persistence of holdings decisions made at bond issuance
This figure shows estimates from the regression specification in Equation (8). The *solid blue line* in panel A plots the estimated pass-through coefficients λ_s for insurer holdings against the quarterly horizon s. For comparison, the *dashed blue line* plots the same regression coefficients using mutual fund and ETF ownership shares in place of the insurer shares in Equation (8). The *solid red line* and *dashed red line* in panel B plot the corresponding R-squared values from these regressions. The *gray-shaded areas* show 95% confidence bands, which use heteroscedasticity-robust standard errors in panel A and are bootstrapped in panel B.

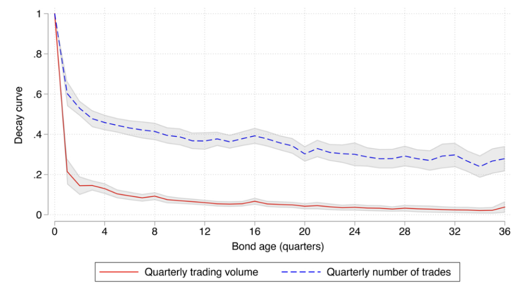

Figure 9
Corporate bond liquidity in the primary market versus the secondary market
This figure shows the estimated coefficients from Equation (9), at varying quarterly horizons. For each horizon, the specification regresses quarterly trading volume (*red solid line*) and quarterly number of trades (*dashed blue line*) for each U.S. investment-grade corporate bond against the same measures in the quarter of bond issuance. The estimated coefficients display a rapid decay toward zero as a consequence of the sharply reduced liquidity of the secondary corporate bond market as compared to the primary market. Standard errors are heteroscedasticity-robust. The *gray-shared areas* correspond to 95% confidence bands.

- Figure 8 shows that insurer purchases at issuance persist for years; Figure 9 shows that liquidity collapses after issuance, so the initial allocation matters.

5.9 Table 4: Shift-share IV

Read:

- The first stage is strong, with F-statistics around 72 to 73.
- The 2SLS estimate of -0.45 is close to the OLS estimate of -0.41.
- That closeness strengthens the causal interpretation rather than weakening it.

5.10 Table 5: Non-U.S. markets

Read:

- Using the complement of fund holdings as a proxy for safe-hand ownership, the paper finds similar negative drawdown elasticities outside the United States.
- The pooled estimate is -0.51, with country-specific estimates of -0.40 for the EMU, -0.64 for the United Kingdom, and -0.35 for Canada.
- This rules out a story that depends on a uniquely U.S. institutional feature.

Table 4
Shift-share IV estimates

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Insurer share	−.40	−.41	−.55	−.45
	(.10)	(.10)	(.15)	(.15)
Fixed effects interaction:				
Event dummy	Yes	Yes	Yes	Yes
Bond size	Yes	Yes	Yes	Yes
Bond duration	Yes	Yes	Yes	Yes
Bond rating	Yes	Yes	Yes	Yes
Issuer dummy	Yes	Yes	Yes	Yes
Callable dummy	—	Yes	—	Yes
Senior dummy	—	Yes	—	Yes
Coupon type	—	Yes	—	Yes
Identifying observations	5,328	4,945	5,328	4,945
Identifying issuers	452	437	452	437
R^2	.84	.85	.84	.85
First-stage F	—	—	71.8	73.3

This table reports the estimates of the shift-share instrumental variable (IV) regression specification in Equation (5) in the sample of actively traded investment-grade U.S. corporate bonds, alongside the baseline OLS estimates (for comparison). The OLS estimates are shown in columns 1 and 2, while the two-stage least squares (2SLS) IV estimates are shown in columns 3 and 4. The various columns correspond to differing sets of bond characteristics included in the interacted fixed effects. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

Table 5
Investor base effects for non-U.S. corporate bonds and proxy approach for U.S. corporate bonds

	Logarithmic bond drawdown				
	(1)	(2)	(3)	(4)	(5)
Proxy safe share	−.39	−.51	−.40	−.64	−.35
	(.14)	(.12)	(.15)	(.21)	(.20)
Sample	USA	Pooled non-U.S.	EMU	GBR	CAN
Fixed effects interaction:					
Event dummy	Yes	Yes	Yes	Yes	Yes
Bond size	Yes	Yes	Yes	Yes	Yes
Bond duration	Yes	Yes	Yes	Yes	Yes
Bond rating	Yes	Yes	Yes	Yes	Yes
Issuer dummy	Yes	Yes	Yes	Yes	Yes
Callable dummy	Yes	Yes	Yes	Yes	Yes
Senior dummy	Yes	Yes	Yes	Yes	Yes
Coupon type	Yes	Yes	Yes	Yes	Yes
Currency FE	—	Yes	Yes	Yes	Yes
Identifying observations	4,945	6,890	4,090	1,484	1,316
Identifying issuers	437	582	393	102	87
R^2	.84	.87	.84	.89	.88

Column 1 reports the estimates of the regression specification in Equation (5), estimated by proxying safe-hand holdings via the complement of fund positions, in the sample of actively traded investment-grade U.S. corporate bonds. Columns 2 through 5 report the estimates of the regression specification in Equation (15). The sample in column 2 pools all investment-grade corporate bonds issued by firms in the European Monetary Union (EMU), United Kingdom, and Canada that are not TRACE-reportable. Columns 3 through 5 show the results separately for each of these three economies. All specifications include fully-saturated interacted fixed effects. The estimates are from weighted least squares (WLS) regressions, with bond sizes as weights. Standard errors are clustered at the issuer by event level and are reported in parentheses.

5.11 Figure 10 and Table 6: Real investment

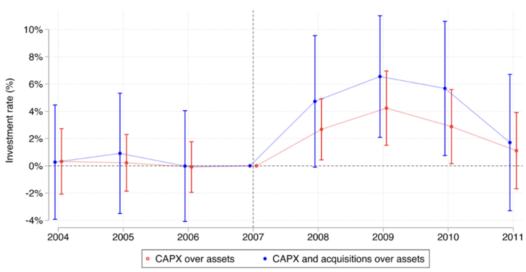

Figure 10
Insurer ownership and firm investment dynamics in the Great Recession
This figure shows the estimated coefficients from the difference-in-differences specification in Equation (18), estimated on the sample of U.S. firms with outstanding bonds in 2007Q1, excluding speculative-grade issuers. The estimates are from just-identified two-stage least squares, using the firm-level instrument in Equation (17), interacted with year dummies. The coefficients quantify the dynamic association between firm investment rates and the share of a firm's bonds held by domestic insurers in 2007Q1. The controls consist of firm-level covariates, measured in the pre-period and interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. The *red line* shows the estimates for capital expenditures (CAPX) over lagged total assets, while the *blue line* shows the estimates for CAPX plus acquisitions over lagged total assets. Standard errors are clustered at the firm level. Bars correspond to 95% confidence intervals.

Table 6
Insurer ownership and investment dynamics in the Great Recession

	Investment rate (%)					
	CAPX			CAPX and acquisitions		
	(1)	(2)	(3)	(4)	(5)	(6)
$\hat{\beta}$	2.80	2.14	1.80	5.42	3.94	2.93
	(1.11)	(.68)	(.69)	(1.73)	(1.04)	(1.08)
$\hat{\eta}$	—	—	1.95	—	—	3.27
	—	—	(.88)	—	—	(1.31)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,994	7,994	7,994	7,994	7,994	7,994
Firms	1,028	1,028	1,028	1,028	1,028	1,028
R^2	.80	.80	.80	.57	.57	.57
Estimator	IV	OLS	OLS	IV	OLS	OLS
First-stage F	65.3	—	—	74.2	—	—

This table reports estimates from the static difference-in-differences specification of Equation (19). The dependent variables in these regressions are firm-level annual investment rates for U.S. firms. The coefficient $\hat{\beta}$ captures the effect of the interaction between the insurer share and the post-period dummy. The coefficient $\hat{\eta}$ captures the effect of the interaction of the insurer share, the post-period dummy, and the share of the firm's outstanding bonds that is due in 2008 or 2009. All regressions include two-way fixed effects for firm and year, as well as controls for firm-level covariates, measured in the pre-period and interacted with the post-period dummy: these are fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. Columns 1 and 4 show estimates from two-stage least squares, using the firm-level instrument in Equation (17), interacted with the post-period dummy. The remaining columns use ordinary least squares. Standard errors are clustered at the firm level and reported in parentheses.

Read:

- Figure 10 shows no meaningful pre-trends before 2007 and then strong positive post-crisis effects of insurer ownership on investment.
- Table 6 shows that the effect is stronger when more debt matures during 2008–2009, exactly as a rollover-risk story would predict.
- The investment effect is quantitatively large and economically plausible when compared with the banking-frictions literature.

5.12 Figure 11: Issuance probability and offering yields

Read:

- Firms with more insurer-held debt are 25 to 30 percentage points more likely to issue new bonds in 2008–2009.

- When they issue, they pay offering yields about 120 basis points lower.
- This is the cleanest evidence that secondary-market fire sales spill over into primary-market financing conditions.

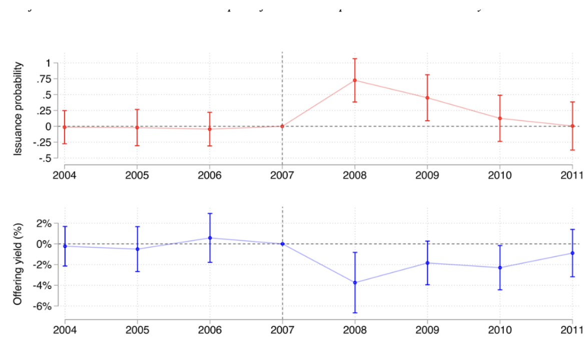

Figure 11
Insurer ownership and firm credit outcomes in the Great Recession
This figure shows the estimated coefficients from the difference-in-differences specification in Equation (20), estimated on the sample of U.S. firms with outstanding bonds in 2007Q1, excluding speculative-grade issuers. The estimates are from just-identified two-stage least squares, using the firm-level instrument in Equation (17), interacted with year dummies. The coefficients quantify the association of yearly firm bond issuance probability (*red line*) and offering yields on new bonds (*blue line*) with the share of a firm's bonds held by domestic insurers in 2007Q1. The controls consist of firm-level covariates, measured in the pre-period and interacted with time dummies: these include fixed effects for issuer rating, average duration of the firm's outstanding bonds, two-digit industry, and deciles of firm size and firm leverage. Standard errors are clustered at the firm level. Bars correspond to 95% confidence intervals.

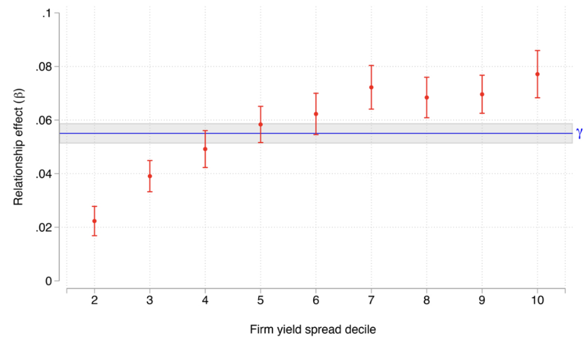

Figure 12
The role of relationships in corporate bond market financing
This figure displays the estimated coefficients from the regression specification in Equation (21). The *solid blue line* displays the coefficient on the relationship indicators $\mathcal{R}_{i,j,k,t}$, which quantifies how much an existing lending relationship increases the probability that a given insurer will buy a firm's new bond offerings for firms in the lowest decile of secondary-market yield spreads. The *red dots* show the estimated coefficients on the interaction between the relationship indicators and firm spread decile dummies, quantifying how this relationship effect varies as firm yield spreads rise. The *red bars* and the *gray-shaded area* plot 95% confidence intervals for these estimates. Standard errors are clustered at the issuer level.

5.13 Figure 12: Relationship lending in bond markets

Read:

- Insurers are already 5.5 percentage points more likely to buy a new issue from a firm they already finance.
- That relationship effect gets much stronger as the firm's outstanding debt trades at wider spreads.
- Safe-handed owners therefore help on two margins: they stabilize old bonds and are also more willing to buy new bonds in bad times.

5.14 Figure 13: Pricing outside crises

Read:

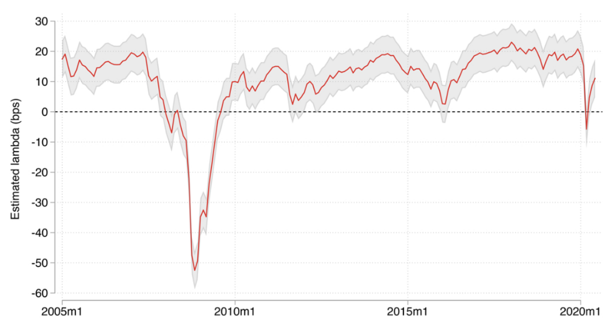

Figure 13
Insurer ownership and yield spread levels
This figure shows the estimates from the regression specification in Equation (22). The *red line* plots the total estimated effect $\Lambda_t = \bar{\Lambda} + \tilde{\Lambda}GZ_t$ of security-level insurer shares on yield spreads to Treasury. The *blue-shaded area* corresponds to a 95% confidence band estimated using Newey and West (1987) HAC standard errors, with lag selection as in Newey and West (1994).

- Outside crises, insurer-held bonds trade at somewhat higher spreads—about 20 basis points from 0 to 1 insurer share.
- In crises, the sign flips and higher insurer ownership becomes valuable.
- The paper interprets this as a liquidity-insurance trade-off: firms pay slightly more in normal times to have a more stable investor base in bad times.

6 Short summary

- **Conclusion.** Who owns a firm's bonds matters for both crisis pricing and real outcomes.
- **Intuition.** Insurers are safe hands because their liabilities are sticky, while mutual funds are weak hands because their liabilities are redeemable.
- **Empirical lesson.** Security-level comparisons within issuer, combined with institutional detail on primary-market allocations, can identify causal ownership effects in asset markets.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Investor composition causally affects corporate bond drawdowns during crises.
- **Magnitude.** A 50-percentage-point increase in insurer ownership makes crisis drawdowns about 20% shallower.
- **Credibility.** The result survives issuer-by-characteristics fixed effects, a strong shift-share IV, extensive robustness checks, and a partial-identification exercise.
- **Mechanism evidence.** Mutual funds face large household-driven redemptions in bad times; insurers do not. Insurers buy primarily at issuance and hold bonds for years.
- **Cross-country relevance.** Similar investor-base effects appear in the EMU, the U.K., and Canada.
- **Real consequence.** Firms financed by stable bondholders borrow more cheaply during crises and invest more.

7.2 Economic intuition

- The paper's key intuition is that ownership structure matters because different intermediaries transform household liabilities differently.
- Open-end funds pass redemption risk through to the assets they hold, which amplifies fire sales exactly when prices are already under pressure.
- Insurers instead act like shock absorbers. Their bonds are less liquid in good times, but precisely that illiquidity is tied to a more stable investor base that becomes valuable in crises.

7.3 Takeaway

This paper shows that the identity of the marginal bondholder is a real state variable for macro-finance. Stable bondholders damp crisis price dislocations, preserve access to primary-market credit, and soften the contraction in firm investment. The broad lesson is that capital allocation in downturns depends not only on borrower fundamentals, but also on the liability structure of the institutions financing those borrowers.